

Roll No.: .....

**END SEMESTER EXAMINATION MAY2025**

Name of the Program: B.Tech

Semester: 8<sup>th</sup>

Name of the Course: Pattern Recognition

Course Code: TCS 852

Time: 3 Hours

Maximum Marks: 100

**Note:**

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

| Q1  | (20Marks)                                                                                                                                                                                               |     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (a) | Construct a confusion matrix for a 3-class classification problem and compute the overall accuracy, precision, and recall for each class.                                                               | CO3 |
| (b) | Compare supervised, unsupervised, and semi-supervised learning approaches in pattern recognition. Discuss their applications with relevant examples.                                                    | CO4 |
| (c) | A researcher uses PCA and t-SNE for visualizing high-dimensional data. Compare these two techniques in terms of dimensionality reduction and interpretability.                                          | CO6 |
| Q2  |                                                                                                                                                                                                         |     |
| (a) | Illustrate the use of normal mixture models in clustering and density estimation. Discuss the key assumptions and advantages these models offer for handling intricate data distributions.              | CO1 |
| (b) | Define the role of similarity measures in clustering. How do they influence clustering outcomes, and what are the standard methods for computing them?                                                  | CO2 |
| (c) | Provide examples of real-world applications of Continuous Hidden Markov Models (HMMs) in areas such as gesture recognition, speech synthesis, and activity recognition.                                 | CO2 |
| Q3  |                                                                                                                                                                                                         |     |
| (a) | Discuss the importance of low-dimensional data representations for visualization and interpretation. What common techniques are used for dimensionality reduction while retaining critical information? | CO2 |
| (b) | Analyze the effect of selecting the number of clusters (K) in K-means clustering. How can one determine the most suitable value of K for a specific dataset?                                            | CO4 |
| (c) | Explain how pattern recognition methods are employed in facial recognition systems and address the key challenges and limitations in their application for security purposes                            | CO6 |
| Q4  |                                                                                                                                                                                                         |     |
| (a) | Explain the state decoding problem in Hidden Markov Models (HMMs) and its practical implications in real-world scenarios.                                                                               | CO1 |
| (b) | Describe how observation densities are estimated in HMMs and explain the roles of the Expectation-Maximization (EM) and Baum-Welch algorithms in this process.                                          | CO2 |
| (c) | Explore the techniques commonly used in handwriting recognition systems and discuss how these methods tackle the variability in handwriting styles.                                                     | CO2 |

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|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Q5</b> |                                                                                                                                                                                                          |     |
| (a)       | Differentiate between the likelihood function and the log-likelihood function in the context of maximum likelihood estimation (MLE).                                                                     | CO4 |
| (b)       | Summarize the three key problems associated with Hidden Markov Models (HMMs) and provide an overview of each.                                                                                            | CO6 |
| (c)       | Highlight the major challenges in designing pattern recognition systems for real-world data. Discuss strategies to address issues such as overfitting, high dimensionality, and variability in datasets. | CO6 |